

# PAPER\_933: Extended AGN Three-Point Jet Power Curves

**Author:** Daniel T. Murphy - Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 212 **Source:** agn\_jet\_power\_curves\_extended.py (ThreePointJetPowerCurve) **Calculator:** ExtendedAGNThreePointJetPowerCalc (CP4 #517) **CVW:** v2.0.0 compliant

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## Abstract

We present explicit three-point jet power enhancement curves for 3C273 and TON618, computed at phonon linewidths  $\Gamma = 0.05, 0.10, \text{ and } 0.30$  THz. Jet modulation via  $M_{\text{jet}}(\Gamma)$  couples to the Blandford-Znajek base power  $P_{\text{BZ}}$  through the relation  $P_{\text{jet}} = P_{\text{BZ}} (1 + M_{\text{jet}})$ . For 3C273 ( $A_{\text{jet}} \sim 1.05$ ), enhancements are 3.1/2.4/1.5x at the three  $\Gamma$  points. For TON618 ( $A_{\text{jet}} \sim 1.40$ ), enhancements reach 3.8/2.9/1.7x, reflecting the stronger phonon-jet coupling in ultra-massive BH systems.

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## 1. Core Equations

Blandford-Znajek power:

$$P_{\text{BZ}} = \frac{B^2}{8\pi} \left( \frac{r_H}{c} \right)^2 a^2 c$$

Jet modulation factor:

$$M_{\text{jet}}(\Gamma) = 1 + A_{\text{jet}} \exp \left( -\frac{(\Gamma - \Gamma_0)^2}{2\sigma_G^2} \right)$$

where  $\Gamma_0 = 2\pi \times 0.1 \times 10^{12}$  rad/s and  $\sigma_G = 0.08 \times 2\pi \times 10^{12}$  rad/s.

Total jet power:

$$P_{\text{jet}} = P_{\text{BZ}} (1 + M_{\text{jet}})$$

Enhancement ratio:

$$\text{Enhancement} = \frac{P_{\text{jet}}(\Gamma)}{P_{\text{BZ}}}$$

## Three-Point Results

| System          | Gamma=0.05 | Gamma=0.10 | Gamma=0.30 |
|-----------------|------------|------------|------------|
| 3C273 (A=1.05)  | 3.1x       | 2.4x       | 1.5x       |
| TON618 (A=1.40) | 3.8x       | 2.9x       | 1.7x       |

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## 2. UQFF Integration

The `ExtendedAGNThreePointJetPowerCalc` (CP4 #517) accepts `M_Msun`, `a_spin`, `B_T`, and `A_jet` as parameters. The `simulate()` method sweeps `A_jet` through [0.8, 1.05, 1.40, 2.0] to map how coupling strength affects the three-point enhancement profile.

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## 3. Source Data

- **File:** `agn_jet_power_curves_extended.py`
  - **Session:** 212
  - **CP4 Class:** `ExtendedAGNThreePointJetPowerCalc` (#517)
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## References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. Blandford, R.D. & Znajek, R.L. – Electromagnetic extraction of energy from Kerr black holes (1977)
3. Ghisellini, G. et al. – General physical properties of bright Fermi blazars (2010)